

R and RStudio Installation

R is a powerful analytical language and contains a number of useful packages for analyzing data.

RStudio is a free and open-source integrated development environment (IDE) for R. RStudio provides comprehensive facilities to R programmers and is highly recommended in this class.

Please complete the following before the start of the class to hit the ground running.

Sections:

- Installing R
- Installing RStudio
- R packages
 - Installing R Packages (Point and Click)
 - Installing R Packages (Code)
 - Exercise: Install an R package

Installing R

We recommend installing the most recent version of R (4.6.0 (called 'Because it was There') as of 2026-04-24). **If you have had installed R already some time ago, we recommend updating/reinstalling it to the most recent version.** Use a link below to launch download of R installers (if the download does not start, a fix may be to copy-paste the below link to your browser):

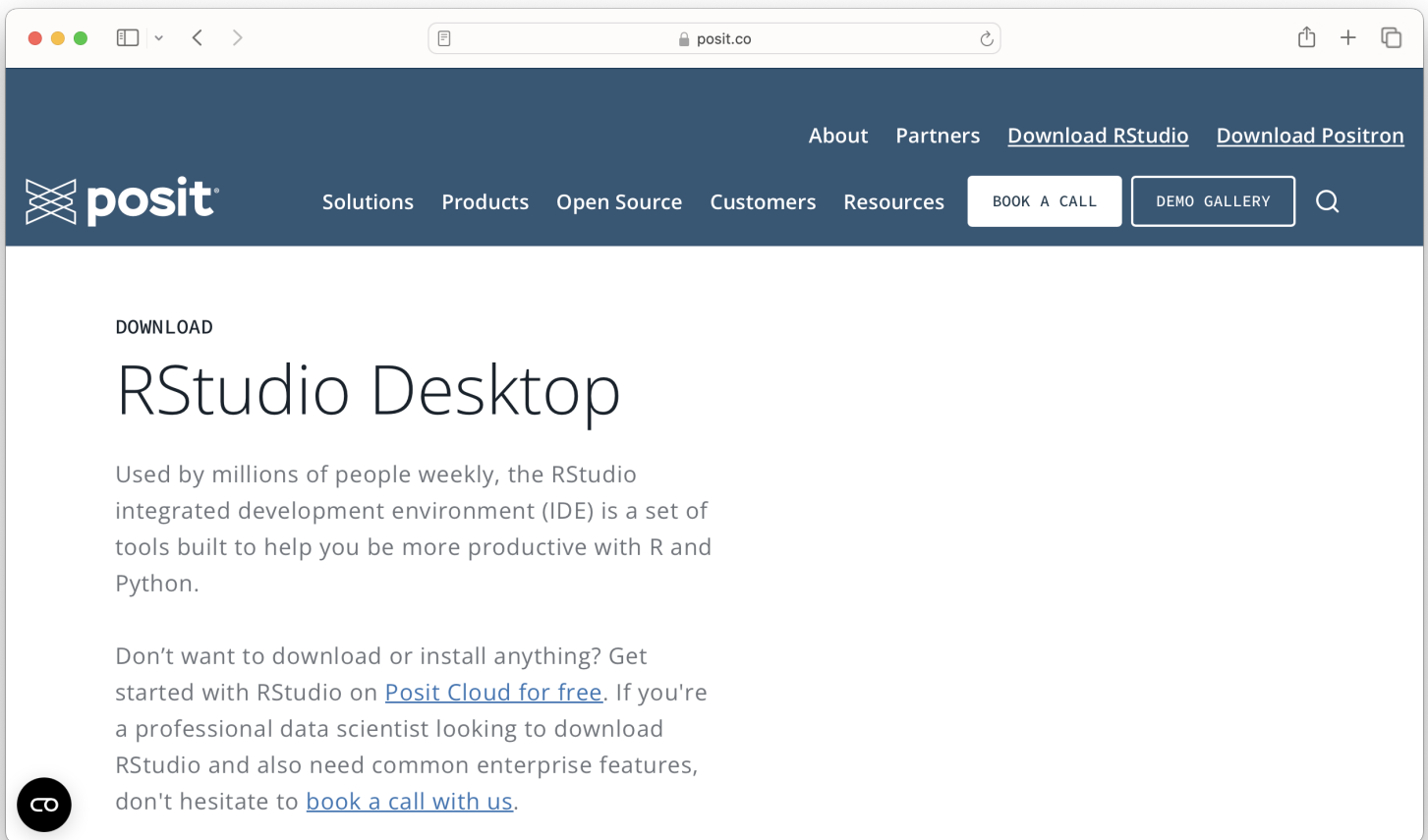
- Download For Mac users (M1 chip) (<https://cran.r-project.org/bin/macosx/sonoma-arm64/base/R-4.6.0-arm64.pkg>)
- Download For Mac users (Intel chip) (https://cran.r-project.org/bin/macosx/big-sur-x86_64/base/R-4.6.0-x86_64.pkg)
- Download For Windows users (<https://cran.r-project.org/bin/windows/base/R-4.6.0-win.exe>)

For other operating systems, or if you prefer to access the download link from the official website, visit <https://cran.r-project.org/> (<https://cran.r-project.org/>) and select Download R for Linux , Download R for macOS or Download R for Windows based on which device you have.

Once the proper installer has been selected, run the installer and follow the on-screen directions. This installation includes the R language and a basic graphical user interface (GUI). Rather using the basic GUI, we recommend installing RStudio - an integrated development environment (IDE) that lets you interact with R with some added benefits.

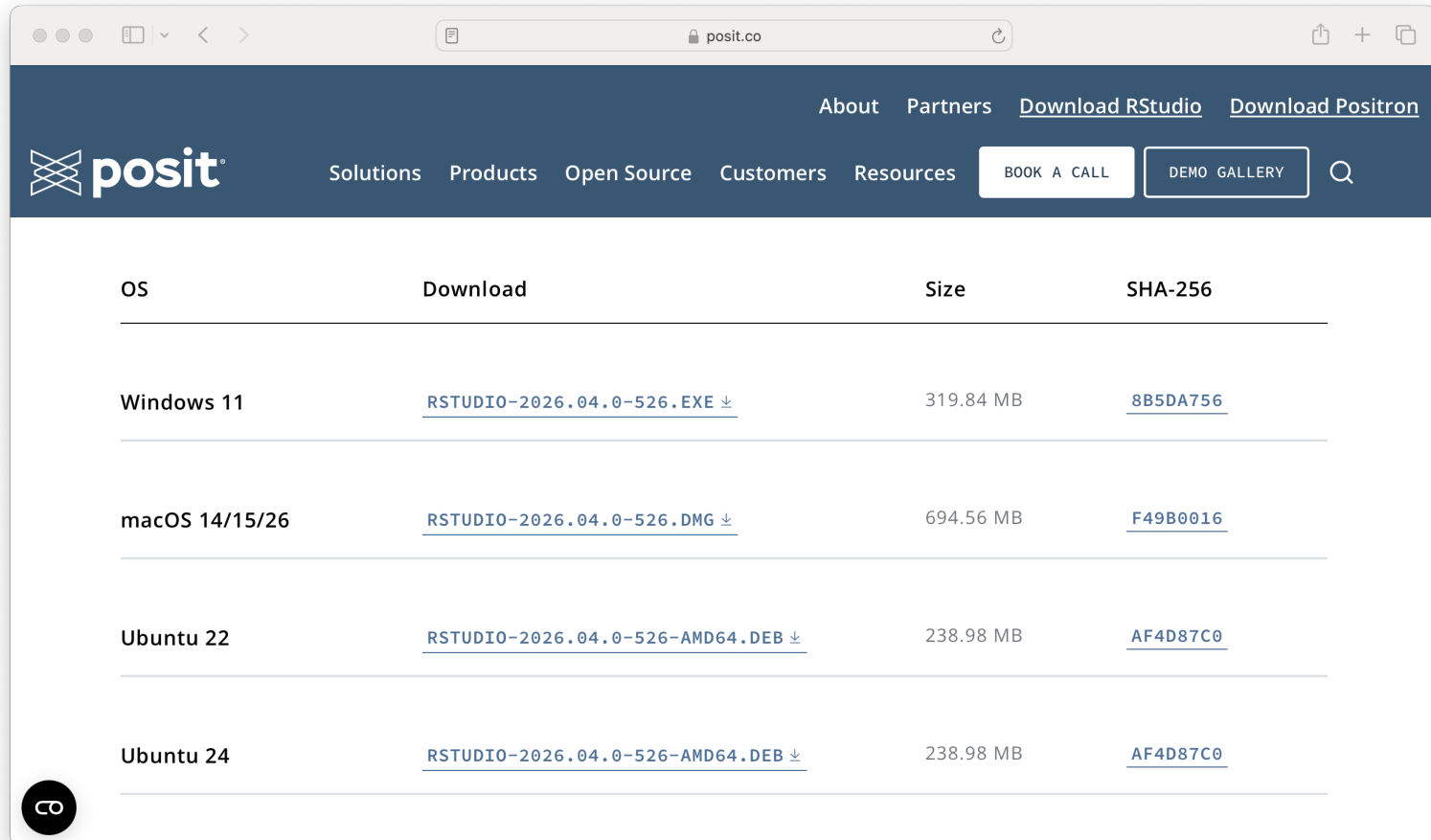
Installing RStudio

To install RStudio , visit <https://posit.co/download/rstudio-desktop/> (<https://posit.co/download/rstudio-desktop/>). The website should look something like this:

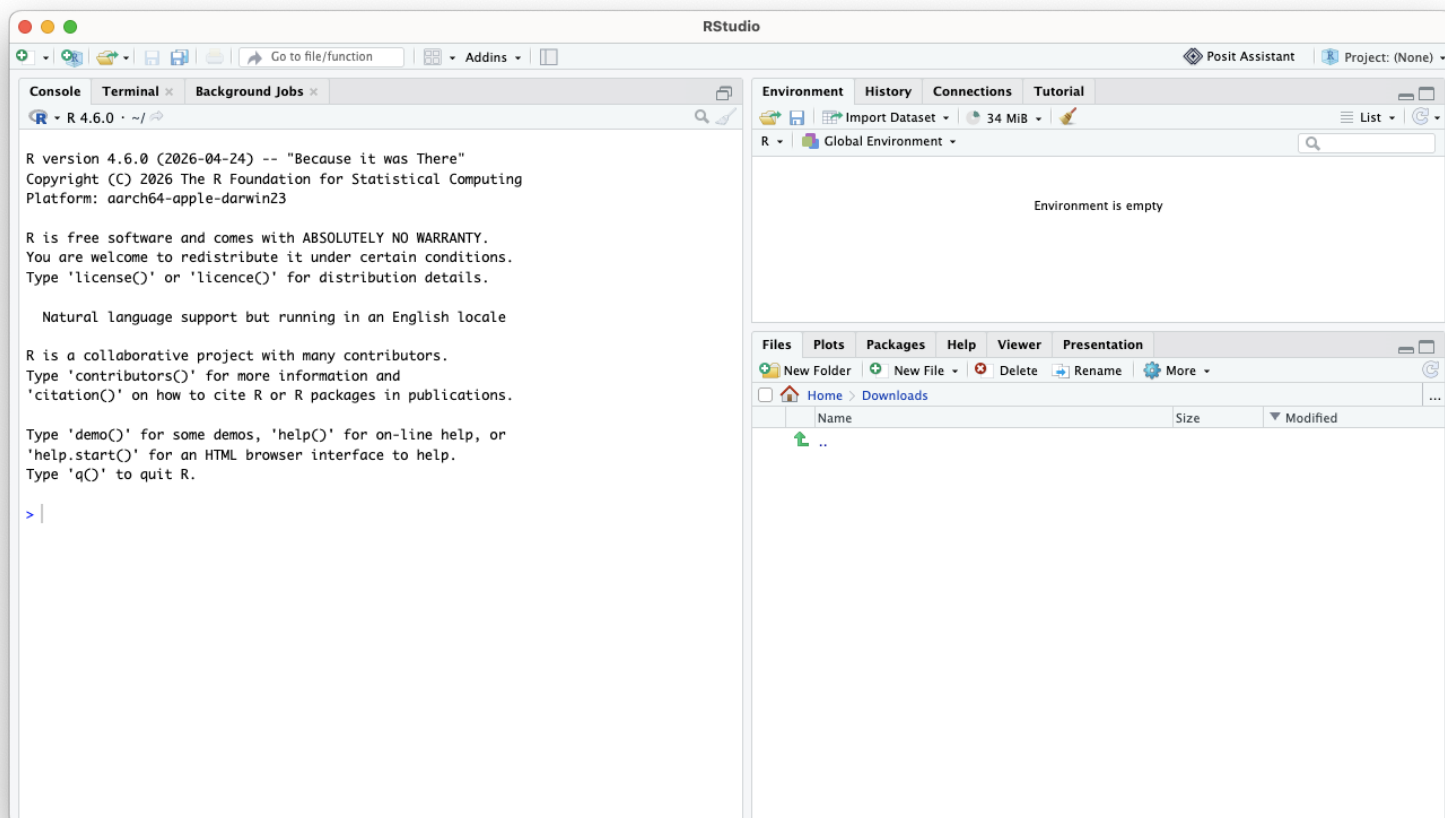


Once on the website, scroll down and select the proper installation file for your platform (Windows, Mac, etc.). Open up the installer and follow the directions to install RStudio.

Note that your operating system may be **too old** to install the current version of RStudio. If this is the case try installing progressively older versions found here (<https://www.rstudio.com/products/rstudio/older-versions/>) until it works. You will know if it worked if you try to open RStudio and you see an interface without a message about things going poorly.



When you open up RStudio , it should look like this:



When you open RStudio for the first time, there are three main windows.

- The **Console** is the left window where you can run lines of code and see the output.

- The **Environment** is the upper-right window where you can manage your data and variables and see previous commands entered (under the “History” tab).
- The **Files** is the bottom-right window where you can look at directories, install packages, see plots, and look at help files for various R commands.

You can customize the look of your RStudio IDE in `Tools > Global Options...`

Troubleshooting the Installation Process

If you run into trouble, did you install the correct version of software for your operating system?

- Check that you installed the version right for your type of system, (macOS vs Windows for example).
- Check if maybe you need a different version for the age of your system. First, check that your version of R was suitable.
- There are multiple versions for different macOS systems. You can check the apple icon (top left corner) and “About This Mac” to learn more about the age of your operating system. If your operating system is older (and you can’t update it), try installing progressively older versions found here (<https://www.rstudio.com/products/rstudio/older-versions/>) until it works. You will know if it worked if you try to open RStudio and you see an interface without a message about things going poorly. Here you can see an example (<https://community.rstudio.com/t/rstudio-desktop-crashes-on-startup-with-library-not-loaded/130296>) of this. Unfortunately, the documentation is poor about what older versions work for which operating systems.
- If you get an error about something called “Rtools”, You probably need some add-on software for a Windows machine. Note that this is not an additional R package to install. Here is a quick guide to installing Rtools: link (https://docs.google.com/viewer?url=https://raw.githubusercontent.com/jhudsl/intro_to_r/main/resources/rtools_windows.pdf).

R packages

Packages are the fundamental units of reproducible R code. They are collections of R code that typically share some common purpose.

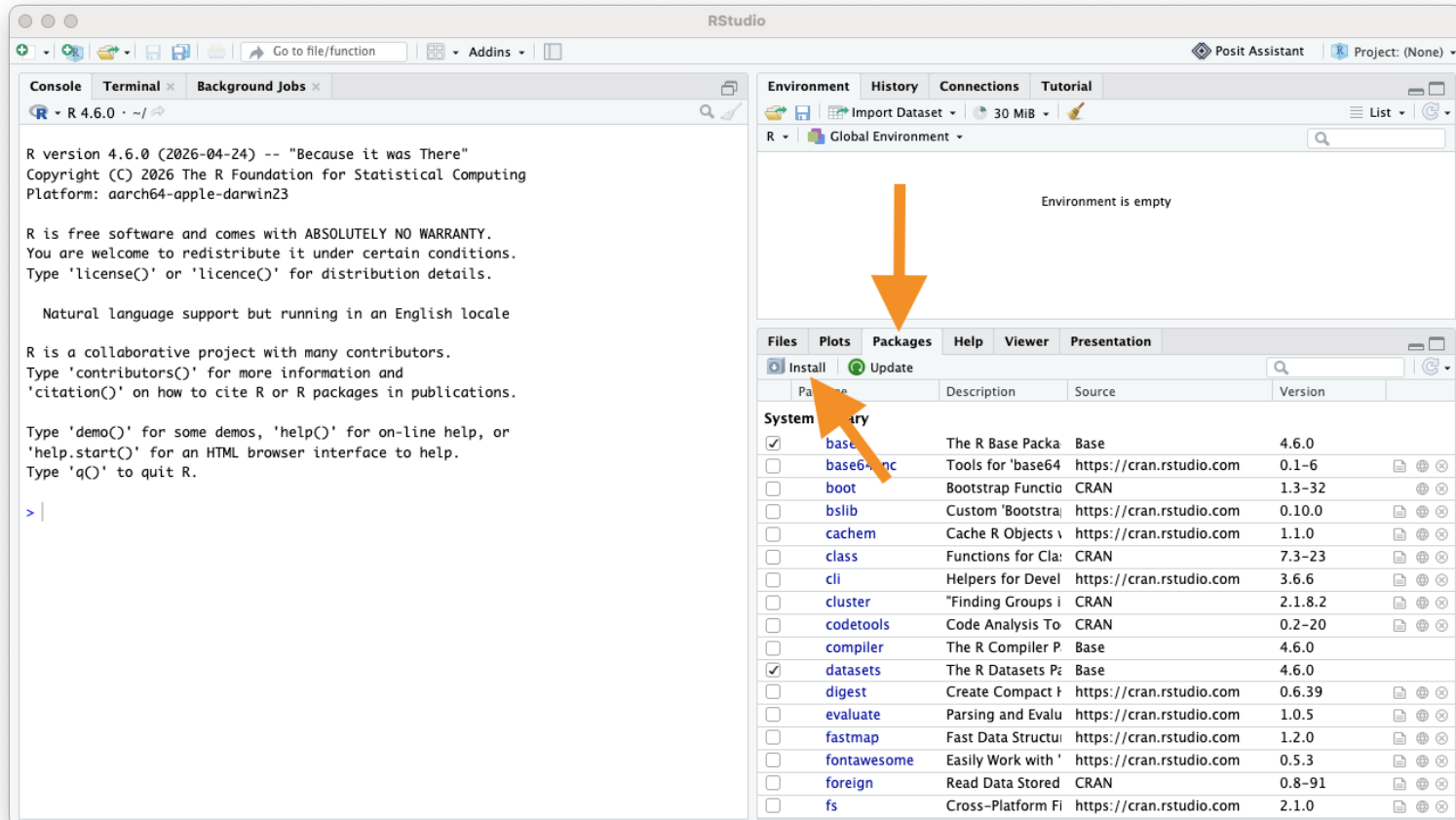
Examples:

- `dplyr` - package of functions for fast data set manipulation (subsetting, summarizing, rearranging, and joining together data sets);
- `ggplot2` - “R’s famous package for making beautiful graphics”; allows to build multiple-layers, highly customizable plots.

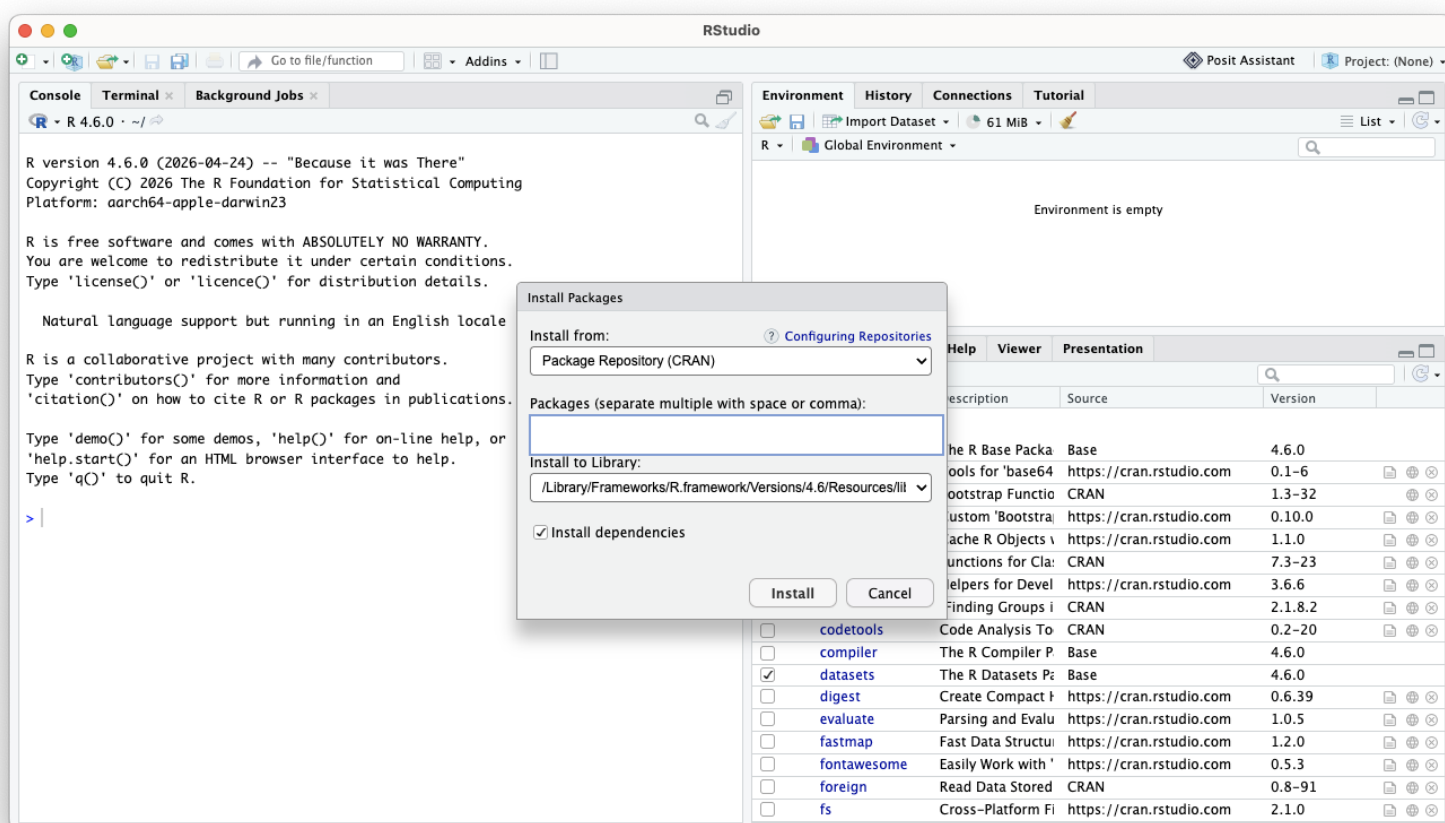
You can install a package via point-and-click methods or with code.

Installing R Package (Point and Click)

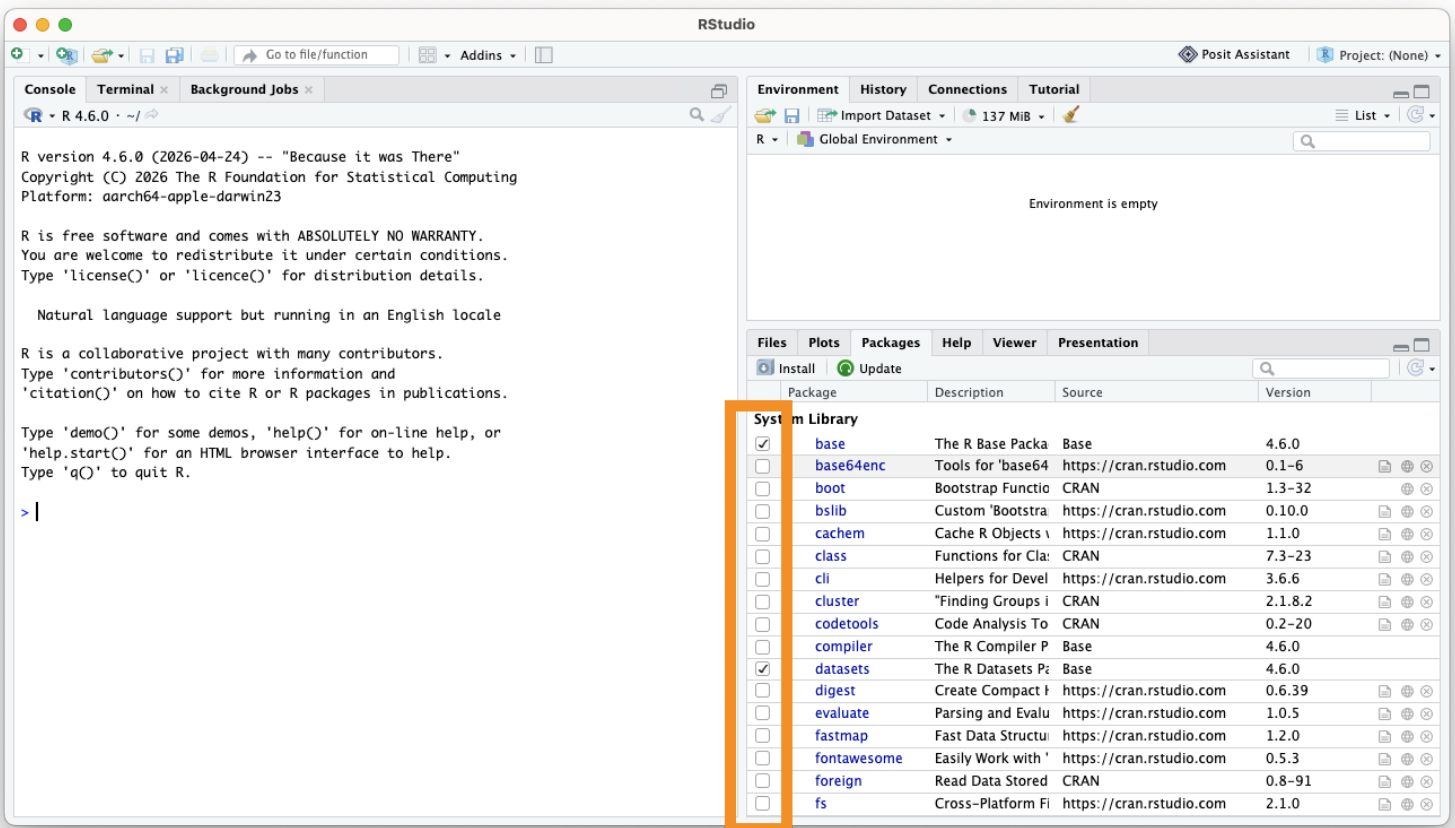
First, click on the “Packages” tab, followed by the “Install” Button.



Select "Package Repository (CRAN)" and type in the name of the package you want to install.



To load the package, check the box next to the package name.



Installing R Package (Code)

To install an R package, type in the RStudio console

```
install.packages("replace_with_package_name")
```

Press enter to execute the command.

Once a package is installed, to use its contents in current R session, we run in the RStudio console the command

```
library(replace_with_package_name)
```

(Note the difference in presence of the quotation mark in the two above commands.)

Exercise: Install an R package

Use the above to install the `tidyverse` package. Execute the `library(...)` command above to check if the package loads successfully.

RStudio

Posit Assistant Project: (None)

Go to file/function Addins

Console Terminal Background Jobs

R - R 4.6.0 - ~/

R version 4.6.0 (2026-04-24) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin23

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

```
> install.packages("tidyverse")
```

Environment History Connections Tutorial

Import Dataset 124 MiB

R Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

Install Update

Package	Description	Source	Version
System Library			
☒ base	The R Base Packa	Base	4.6.0
☐ base64enc	Tools for 'base64	https://cran.rstudio.com	0.1-6
☐ boot	Bootstrap Functio	CRAN	1.3-32
☐ bslib	Custom 'Bootstra	https://cran.rstudio.com	0.10.0
☐ cachem	Cache R Objects i	https://cran.rstudio.com	1.1.0
☐ class	Functions for Cla	CRAN	7.3-23
☐ cli	Helpers for Devel	https://cran.rstudio.com	3.6.6
☐ cluster	"Finding Groups i	CRAN	2.1.8.2
☐ codetools	Code Analysis To	CRAN	0.2-20
☐ compiler	The R Compiler P	Base	4.6.0
☒ datasets	The R Datasets P	Base	4.6.0
☐ digest	Create Compact I	https://cran.rstudio.com	0.6.39
☐ evaluate	Parsing and Evalu	https://cran.rstudio.com	1.0.5
☐ fastmap	Fast Data Structu	https://cran.rstudio.com	1.2.0
☐ fontawesome	Easily Work with	https://cran.rstudio.com	0.5.3
☐ foreign	Read Data Stored	CRAN	0.8-91
☐ fs	Cross-Platform Fi	https://cran.rstudio.com	2.1.0

RStudio

Posit Assistant Project: (None)

Go to file/function Addins

Console Terminal Background Jobs

R - R 4.6.0 - ~/

```
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/rapp1.4.0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/lubridate_1.9.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/magrittr_2.0.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/modelr_0.1.11.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/pillar_1.11.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/purrr_1.2.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/ragg_1.5.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/readr_2.2.0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/readxl_1.4.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/reprex_2.1.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/rstudioapi_0.18.0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/rvest_1.0.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/stringr_1.6.0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/tibble_3.3.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/tidyr_1.3.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/xml2_1.5.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/tidyverse_2.0.0.tgz'
```

The downloaded binary packages are in
/var/folders/qz/3111t6y102gbbth2vmm_2ch0000gn/T//RtmpFpdoSz/downloaded_packages

```
> library(tidyverse)
```

Environment History Connections Tutorial

Import Dataset 206 MiB

R Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

Install Update

Package	Description	Source	Version
System Library			
☐ askpass	Password Entry U	https://cran.rstudio.com	1.2.1
☐ backports	Reimplementation	https://cran.rstudio.com	1.5.1
☒ base	The R Base Packa	Base	4.6.0
☐ base64enc	Tools for 'base64	https://cran.rstudio.com	0.1-6
☐ bit	Classes and Meth	https://cran.rstudio.com	4.6.0
☐ bit64	A S3 Class for Ve	https://cran.rstudio.com	4.8.0
☐ blob	A Simple S3 Class	https://cran.rstudio.com	1.3.0
☐ boot	Bootstrap Functio	CRAN	1.3-32
☐ broom	Convert Statistica	https://cran.rstudio.com	1.0.13
☐ bslib	Custom 'Bootstra	https://cran.rstudio.com	0.10.0
☐ cachem	Cache R Objects i	https://cran.rstudio.com	1.1.0
☐ callr	Call R from R	https://cran.rstudio.com	3.7.6
☐ cellranger	Translate Spread	https://cran.rstudio.com	1.1.0
☐ class	Functions for Cla	CRAN	7.3-23
☐ cli	Helpers for Devel	https://cran.rstudio.com	3.6.6
☐ clipr	Read and Write fr	https://cran.rstudio.com	0.8.0
☐ cluster	"Finding Groups i	CRAN	2.1.8.2

The screenshot shows the RStudio interface with the following components:

- Console:** Displays the output of the `install.packages()` command. It shows the download of binary packages for macOS ARM64 and the installation of the tidyverse suite. The output includes the following table of installed packages:

Package	Version
dplyr	1.2.1
readr	2.2.0
forcats	1.0.1
stringr	1.6.0
ggplot2	4.0.3
tibble	3.3.1
lubridate	1.9.5
tidyr	1.3.2
purrr	1.2.2

Below the table, it shows conflicts with the `stats` package:

```

Conflicts:
  * dplyr::filter() masks stats::filter()
  * dplyr::lag() masks stats::lag()
Use the conflicted package to force all conflicts to become errors

```

- Environment:** Shows the Global Environment, which is currently empty.
- Packages:** Lists the installed packages in the system library, including askpass, backports, base, base64enc, bit, blob, boot, broom, bslib, cachem, callr, cellranger, class, cli, clipr, and cluster.

If you want extra practice, we will also be using the packages `rmarkdown`, `nanianr`, and `janitor`.